FAIR-SPLIT: NEAR-OPTIMAL DECISION TREES FOR ACCURACY, SPARSITY, AND FAIRNESS

Ziyi Cheng
College of Rixin
Tsinghua University
zy-cheng24@mails.tsinghua.edu.cn

Kaidi Zha
Department of Physics
Tsinghua University
ckd23@mails.tsinghua.edu.cn

ABSTRACT

Decision trees are widely adopted for their computational efficiency and interpretability. However, greedy induction algorithms, such as CART, are prone to local optima, and exhaustive global search strategies are prohibitively expensive for large-scale data. Moreover, feature bias inherent in certain datasets can degrade predictive performance and fairness. Building on the SPLIT family of algorithms, we study a hybrid search framework that performs optimal lookahead over a shallow prefix of the tree while recursively completing the remaining subtrees. This design strikes a favorable trade-off between global optimality and computational tractability, producing sparse trees that are amenable to fairness auditing. Across five random seeds on 12 benchmark datasets, SPLIT-family models are competitive with CART on several tasks and outperform it on selected datasets such as COMPAS, HELOC, Heart, and Law School, while CART remains strongest on many standard accuracy benchmarks. For group fairness, removing sensitive attributes before training is not sufficient on its own; label-aware sample-level post-processing sharply reduces demographic-parity gaps, while Leaf-Pareto Fair Recalibration (LPFR) provides a more conservative deployable leaf-level adjustment. Our code is publicly available in https://github.com/illusionsin30/fair-SPLIT.

1 INTRODUCTION

Decision trees (Breiman et al., 1984) are among the most widely deployed machine learning models, owing to their computational efficiency, intuitive interpretability, and broad applicability. Their transparent structure—each prediction traceable through a short sequence of human-readable feature splits—makes them especially valuable in high-stakes domains such as healthcare, criminal justice, and credit lending, where regulatory and ethical demands increasingly require decisions to be explainable, auditable, and free from discrimination (Rudin, 2019).

However, standard tree induction algorithms suffer from two fundamental shortcomings. First, greedy heuristics such as CART and C4.5 make myopic split choices that optimize local impurity reduction but offer no guarantee on the global tree structure, often yielding suboptimal trees (Murthy et al., 1994). Second, greedy algorithms exhibit well-documented feature selection bias—favoring features with many distinct values—which can amplify discrimination against protected subgroups (Deng et al., 2011; Strobl et al., 2007). While globally optimal tree induction can mitigate both issues, the problem is NP-hard (Hyafil & Rivest, 1976), and exact solvers such as GOSDT (Lin et al., 2020) and MurTree (Demirović et al., 2022) do not scale to large datasets.

The recently proposed SPLIT framework (Babbar et al., 2025) offers a principled middle ground: it performs an exact dynamic programming search over the top d_l layers of the tree (the *prefix*), where splits carry the highest discriminative leverage, and fills the remaining leaves greedily or optimally. This hybrid design yields near-optimal trees at a fraction of the cost of full exhaustive search. The SPLIT objective directly penalizes tree size through a sparsity regularizer λ, producing compact trees amenable to fairness auditing. Two extensions further enrich the framework: LicketySPLIT, a polynomial-time variant, and ReSPLIT, which samples diverse near-optimal trees from the Rashomon set (Breiman, 2001; Semenova et al., 2022).

In this work, we present fair-SPLIT, a pipeline that combines the SPLIT family with lightweight fairness interventions for studying accuracy, sparsity, and demographic parity. Our contributions are threefold. First, we integrate a fairness-aware pre-processing step that removes sensitive attributes before training, preventing direct discrimination while preserving non-sensitive features. Second, we compare two post-training fairness interventions: a label-aware sample-level adjustment that diagnoses how much demographic-parity gap can be removed by targeted hard-prediction flips, and Leaf-Pareto Fair Recalibration (LPFR), which keeps the learned tree structure fixed and learns leaf-level prediction overrides on a calibration split. Third, using five-seed summary results on 12 benchmark datasets, we show that no single tree learner dominates: CART is strongest on many conventional accuracy benchmarks, while SPLIT-optimal, SPLIT-greedy, and ReSPLIT each lead on selected datasets. Fairness improvements are similarly method- and dataset-dependent, with sample-level calibration nearly eliminating SP but LPFR providing the more deployable post-processing mechanism. Our code is publicly available at `https://github.com/illusionsin30/fair-SPLIT`.

2 Related Work

2.1 Optimal and Sparse Decision Trees

The problem of constructing optimal binary decision trees is NP-complete (Hyafil & Rivest, 1976). Early exact approaches used linear programming (Verwer & Zhang, 2019) and constraint programming (Aglin et al., 2020). More recently, dynamic programming with branch-and-bound has become the dominant paradigm: OSDT (Hu et al., 2019) introduced the core DP formulation, GOSDT (Lin et al., 2020) added threshold guessing and improved pruning, and MurTree (Demirović et al., 2022) further tightened the bounds. Recent work has continued to broaden this space: van der Linden et al. (2023) characterize when dynamic programming objectives decompose cleanly over subtrees, Chaouki et al. (2025) formulate sparse tree search as an AND/OR graph solved by an AO*-style algorithm, and Kohler et al. (2025) recast non-greedy tree induction as an MDP. These solvers improve search quality but their depth and feature dependence still limits scalability. SPLIT (Babbar et al., 2025) addresses this through a hybrid strategy: exact optimization of the top few layers via DP, combined with greedy completion of the remaining leaves. The search space is controlled through feature binarization—using midpoint thresholds or GBDT stump-based threshold guessing (McTavish et al., 2022)—which is essential for tractable DP state enumeration.

Another line of work replaces discrete split search with differentiable or neuralized tree models. For example, Yang et al. (2018) realize decision trees as neural networks that can be trained by gradient descent. These methods are useful for integrating tree structure into neural toolchains, but they do not provide the same sparse optimality objective as SPLIT-family methods. Our focus is therefore on sparse axis-aligned trees whose discrete structure remains directly auditable.

2.2 Fairness in Machine Learning and Decision Trees

Fairness in machine learning has been formalized through several families of definitions (Barocas et al., 2017; Mehrabi et al., 2021). Group fairness criteria include demographic parity (equal positive prediction rates across groups), equalized odds, and equality of opportunity (Hardt et al., 2016). Interventions fall into three paradigms: pre-processing, which modifies training data (Zemel et al., 2013; Kamiran & Calders, 2012); in-processing, which incorporates fairness constraints into the objective (Agarwal et al., 2018; Zafar et al., 2019); and post-processing, which adjusts model outputs (Hardt et al., 2016; Pleiss et al., 2017). Friedler et al. (2019) showed that no single intervention dominates across all settings. Menon & Williamson (2018) proved that when fairness violations arise primarily from unequal base rates, post-processing can achieve near-zero fairness gaps with negligible accuracy loss.

Decision trees have attracted particular attention because their transparent structure facilitates auditing and their piecewise-constant predictions align naturally with group-specific threshold adjustments. Early work modified the splitting criterion to jointly consider accuracy and fairness (Kamiran & Calders, 2009), while Fair Forests regularize tree induction to reduce model bias (Raff et al., 2018). Subsequent approaches incorporated fairness constraints into MIP formulations for fair and optimal classification trees (Aghaei et al., 2019; Jo et al., 2023). Related tree-based fairness work also includes missing-value-aware fair trees (Jeong et al., 2022), fairness-constrained gradient boosting (Cruz et al., 2023), and online oblique decision forests for group fairness (Chowdhury et al., 2024). The Rashomon set perspective—enumerating many near-optimal trees and selecting the fairest—has recently emerged as a promising direction (Xin et al., 2022; Donnelly et al., 2023), discovering fair models within the set of high-accuracy solutions without explicit fairness constraints.

Post-processing for fairness typically adjusts per-group decision thresholds. Hardt et al. (2016) proposed group-specific ROC curves to achieve equalized odds. Pleiss et al. (2017) established that perfect calibration and equalized odds cannot simultaneously hold unless base rates are equal across groups. This tension motivates designs that prioritize one fairness criterion—in our case, demographic parity—while monitoring accuracy damage. The sample-level procedure used in our experiments follows the hard-prediction flipping intuition behind threshold post-processing, with the additional principle of prioritizing already-misclassified examples when labels are available. LPFR instead operates at the leaf level, making it closer to a deployable post-processing rule for tree models.

3 Preliminaries

A decision tree T partitions the input space $\mathcal{X} \subset \mathbb{R}^d$ into disjoint leaf regions $R_1, \ldots, R_M$. Each internal node is associated with a split $s = (j, \tau)$, where $j \in \{1, \ldots, d\}$ is the feature index and $\tau \in \mathbb{R}$ is a threshold; the left and right children receive subsets $\mathcal{D}_L = \{(x, y) \mid x_j \leq \tau\}$ and $\mathcal{D}_R = \mathcal{D} \setminus \mathcal{D}_L$.

The CART algorithm (Breiman et al., 1984) is the canonical greedy tree induction method. At each internal node, CART selects the split that minimizes the weighted Gini impurity of the resulting child nodes, where the Gini index measures class heterogeneity within a dataset (formal definitions are provided in Appendix A). This recursive partitioning continues until a stopping criterion is met.

Despite its popularity, CART suffers from two well-documented limitations. First, its myopic split selection is prone to local optima (Murthy et al., 1994). Second, the exhaustive search over all features introduces selection bias favoring features with many distinct values or high variance, which can degrade both accuracy and fairness (Deng et al., 2011; Strobl et al., 2007). Global optimization over the entire tree mitigates these issues but is NP-hard (Hyafil & Rivest, 1976).

3.1 Regularized Tree Learning and Binarization

The SPLIT framework (Babbar et al., 2025) formulates tree learning as a regularized risk minimization problem. Let T be a decision tree with leaf set $\mathcal{L}(T)$. Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, the objective is

$$L(T, \mathcal{D}, \lambda) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[y_i \neq T(x_i)] + \lambda \cdot |\mathcal{L}(T)|, \tag{1}$$

where $|\mathcal{L}(T)|$ denotes the number of leaves and $\lambda \geq 0$ controls the accuracy–sparsity tradeoff. When $\lambda = 0$, the problem reduces to unregularized empirical risk minimization; as $\lambda \to \infty$, the optimal tree is a single leaf. Sweeping λ traces the full Pareto frontier between error and complexity.

The exact DP solver in SPLIT operates on binary features to keep the state space tractable. Two strategies are implemented. Midpoint binarization creates binary columns from candidate thresholds and can produce many columns for high-cardinality features. Threshold guessing (McTavish et al., 2022) trains a GBDT ensemble of depth-1 stumps and collects their split thresholds as candidate binary features, yielding a compact feature set. Binarized datasets are denoted $\tilde{X} \in \{0, 1\}^{N \times p}$.

3.2 Fairness Evaluation

We evaluate group fairness using three standard metrics (Barocas et al., 2017; Hardt et al., 2016; Dwork et al., 2012), computed on a held-out test set from hard predictions $\hat{Y} \in \{0, 1\}$ given protected attribute A partitioning the data into groups G. Statistical Parity Difference (SPD) measures the maximum gap in positive prediction rates across groups; an SPD of 0 indicates perfect demographic parity, and values above 0.1 are conventionally considered unfair. Disparate Impact (DI) is the ratio of the smallest to the largest group positive rate, with ideal value 1; the four-fifths rule considers DI < 0.8 to indicate adverse impact. Equal Opportunity Difference (EOD) captures the maximum gap in true positive rates across groups; an EOD of 0 indicates that all groups have equal opportunity of being correctly classified as positive. Formal mathematical definitions are given in Appendix A. Groups with fewer than 50 test samples are excluded due to statistical unreliability.

4 FAIR-SPLIT

We now describe fair-SPLIT, which integrates SPLIT tree learning with fairness-aware preprocessing and post-training calibration.

4.1 THE SPLIT ALGORITHM

SPLIT (SParse Lookahead for Interpretable Trees) is a two-phase hybrid algorithm. The key insight is that the most discriminative splits occur near the root, where data partitions are largest; deeper splits operate on smaller, increasingly homogeneous subsets where greedy choices suffice.

Phase 1: Optimal Prefix Search. Let d be the total depth budget and $d_l < d$ the lookahead depth. Phase 1 constructs an optimal prefix of depth d_l minimizing Equation 1 over all depth-d_l trees, via DP with memoization. Let the DP state be (X, y, δ) where δ is the remaining depth. The recursion is

$$\mathrm{DP}(X, y, \delta) = \min\left\{ \underbrace{\lambda + \frac{1}{N}\sum\nolimits_{i:y_i \neq \hat{y}_{\text{maj}}} 1}_{\text{leaf}},\ \min_{j\in\mathcal{F}} \left\{\mathrm{DP}(X_L^j, y_L^j, \delta{-}1) {+} \mathrm{DP}(X_R^j, y_R^j, \delta{-}1)\right\} \right\}, \quad (2)$$

where $\mathcal{F}$ is the set of binary candidate features and $\hat{y}_{\text{maj}}$ denotes the majority class label of the data subset at the current node, so the leaf loss is the fraction of samples whose labels differ from the majority label. The search space grows as $O(k^\delta)$. Three accelerations are used: memoization via hash of label vector and depth; optional feature pre-filtering to a top-k subset ranked by entropy gain; and branch-and-bound using greedy completion as an upper bound at the lookahead boundary. Phase 1 outputs leaf subsets (X_ℓ, y_ℓ) at depth d_l.

Phase 2: Leaf Filling. Each leaf is completed to depth $d - d_l$. Greedy fill uses λ-gated split acceptance (Algorithm 1): a split on feature j is accepted only if the sum of regularized child losses is strictly less than the leaf loss. Optimal fill re-invokes the DP solver per leaf. The total time complexity is $O(n \cdot k^{d_l+1} + n \cdot k^{d-d_l})$.

4.2 LICKETYSPLIT AND RESPLIT

LicketySPLIT provides a polynomial-time alternative when k or d makes the exponential DP prohibitive. It restricts the lookahead to $d_l = 1$: at each node, every candidate feature is evaluated by greedily completing both children, and the best root split is selected, recursing on both children. Formally,

$$j^* = \arg\min_{j\in\mathcal{F}} \left\{ L_{\text{greedy}}\big(X_L^j, y_L^j, \delta - 1, \lambda\big) \ + \ L_{\text{greedy}}\big(X_R^j, y_R^j, \delta - 1, \lambda\big) \right\}, \quad (3)$$

with time complexity $O(n \cdot k^2 \cdot d^2)$ (Babbar et al., 2025).

ReSPLIT constructs an ε-Rashomon set $\mathcal{R}(\varepsilon) = \{T : L(T, \mathcal{D}, \lambda) \leq (1 + \varepsilon) \cdot L^*\}$ (Breiman, 2001; Semenova et al., 2022) through three steps: generate N diverse prefixes by shuffling feature order and using the `pick_first` strategy; greedily or optimally complete each prefix into a full tree; and discard trees exceeding the ε bound. Following the SPLIT framework's design principle of minimizing the regularized objective, the reported ReSPLIT prediction uses the tree with the lowest regularized risk in the retained set; fairness-aware model selection from the Rashomon set is left as a downstream step that can be composed with LPFR or other post-processing.

Algorithm 1 Greedy Tree Builder

Require: X, y, depth budget d, regularization λ
Ensure: A decision tree of depth $\leq d$
if $d = 0$ or y is pure **then**
    return LEAF(majority(y))
end if
lb_leaf $\leftarrow \lambda + \text{wrong}(y, \text{majority}(y))/N$
Rank features by entropy gain; select top-k candidates
for each candidate feature j in ranked order **do**
    $(X_L, y_L), (X_R, y_R) \leftarrow \text{split}(X, y, j)$
    $\text{lb}_L \leftarrow \text{GREEDY}(X_L, y_L, d-1, \lambda)$
    $\text{lb}_R \leftarrow \text{GREEDY}(X_R, y_R, d-1, \lambda)$
    if $\text{lb}_L + \text{lb}_R <$ lb_leaf **then**
        return NODE$(j,\ \text{GREEDY}(X_L, y_L, d-1, \lambda),\ \text{GREEDY}(X_R, y_R, d-1, \lambda))$
    end if
end for
return LEAF(majority(y))

4.3 Fairness-Aware Pre-Processing and Post-Training Calibration

The first fairness intervention removes all sensitive attribute columns (e.g., `sex`, `race`) from the feature matrix before training, eliminating direct use of the protected attribute. Training is performed on $X_{\text{fair}} = X \setminus A$, preserving all non-sensitive features. This step incurs negligible computational overhead and is compatible with any downstream algorithm. Indirect discrimination via proxy features is not automatically removed; the remaining fairness behavior is therefore evaluated empirically.

The second component is a sample-level post-training adjustment that targets demographic parity by modifying hard predictions within each group. Given predictions $\hat{Y}$, labels Y, and sensitive attribute A, the global target rate $\bar{p} = \frac{1}{N}\sum_i \hat{y}_i$ is computed. For each group g with positive rate $\hat{p}_g$, if $\hat{p}_g > \bar{p}$, then predictions are flipped from 1 to 0 until the group positive count approaches $\bar{p}n_g$, prioritizing false positives (where $Y = 0$) to correct errors. If $\hat{p}_g < \bar{p}$, 0-to-1 flips are applied, prioritizing false negatives. Random tie-breaking handles cases where priority-correctable examples are insufficient. This algorithm is model-agnostic and runs in $O(N)$ time with no retraining, but because it uses ground-truth labels to choose flips, the calibrated accuracy results should be read as diagnostic post-processing results rather than a deployable inference-time rule.

The latest fair-SPLIT implementation also includes Leaf-Pareto Fair Recalibration (LPFR). LPFR freezes the trained tree structure, identifies leaves on a calibration split, and iteratively selects leaf-level prediction overrides that improve fairness while respecting a user-specified tolerance on accuracy drop. At each step, LPFR picks the leaf flip with the best fairness-gain-to-accuracy-loss ratio, continuing until any further fairer overrides would exceed the allowed accuracy budget. Unlike sample-level flipping, LPFR can be applied at inference time using only the sample's path through the tree. We therefore report sample-level calibration as an optimistic diagnostic baseline and LPFR as the deployable recalibration variant.

The main mechanisms of fair-SPLIT—sparsity regularization (λ), ReSPLIT diversity (ε), sample-level calibration, and LPFR—are orthogonal and composable. The full experimental pipeline proceeds: drop sensitive attributes when fairness mode is enabled, train with CART/SPLIT/ReSPLIT, and optionally apply one of the post-processing methods. This layered design lets practitioners explore the accuracy–sparsity–fairness Pareto surface without expensive constrained optimization.

5 Experiments

We evaluate on 12 benchmark datasets across three dimensions: predictive accuracy, training efficiency, and group fairness, comparing CART, SPLIT-greedy, SPLIT-optimal, and ReSPLIT. Unless otherwise noted, reported numbers are means and standard deviations over five random seeds from `summary.csv`.

Table 1: Test accuracy in nofair mode (%, mean±std over five random seeds) on the 12 benchmark datasets. The best mean accuracy for each dataset is highlighted in **bold**.

Algorithms	Adult	Bank	COMPAS	German
CART	**84.90±0.38**	**90.08±0.51**	66.10±0.78	**72.40±1.98**
SPLIT-greedy	84.05±0.26	89.79±0.56	65.96±0.72	71.00±2.47
SPLIT-optimal	84.69±0.23	89.17±0.92	**66.38±0.53**	70.30±2.08
ReSPLIT	83.83±0.33	88.35±1.16	66.10±1.19	69.40±2.68

Algorithms	HELOC	Spambase	Covertype	Thyroid
CART	68.14±0.80	**90.73±0.93**	**70.16±0.19**	**93.60±0.48**
SPLIT-greedy	69.38±2.05	89.21±0.59	66.93±0.13	89.82±1.15
SPLIT-optimal	69.22±2.43	90.18±0.85	67.17±0.21	90.02±0.85
ReSPLIT	**70.26±2.31**	89.56±1.28	67.26±0.32	89.48±0.68

Algorithms	Communities	Heart	Law School	ACSIncome
CART	**96.13±0.59**	76.39±2.49	94.93±0.14	**78.76±0.15**
SPLIT-greedy	94.57±0.87	75.08±3.74	**95.02±0.07**	75.48±0.42
SPLIT-optimal	95.33±0.95	76.72±5.73	94.98±0.14	75.84±0.35
ReSPLIT	95.28±1.15	**78.69±2.32**	94.90±0.04	76.10±0.32

Table 2: Training time in nofair mode (seconds, mean±std over five random seeds). The fastest mean time for each dataset is highlighted in **bold**.

Algorithms	Adult	Bank	COMPAS	German
CART	6.637±3.168	1.125±0.201	1.318±0.290	0.436±0.045
SPLIT-greedy	**6.283±4.951**	**0.718±0.142**	**1.318±1.236**	**0.403±0.136**
SPLIT-optimal	16.176±5.152	1.157±0.277	1.546±1.030	0.909±0.847
ReSPLIT	1743.5±28.6	2.320±0.502	30.036±13.669	6.310±8.010

Algorithms	HELOC	Spambase	Covertype	Thyroid
CART	1.164±0.160	3.898±1.478	**83.266±18.460**	2.909±0.603
SPLIT-greedy	**0.695±0.342**	**1.950±1.146**	250.5±20.4	**0.886±0.115**
SPLIT-optimal	0.844±0.221	2.500±1.150	878.8±866.8	1.080±0.257
ReSPLIT	7.066±2.194	32.335±7.331	2716.1±134.0	10.592±6.401

Algorithms	Communities	Heart	Law School	ACSIncome
CART	11.690±3.986	**0.132±0.039**	**4.796±1.757**	**12.119±6.271**
SPLIT-greedy	**2.794±0.853**	0.343±0.328	5.646±6.522	13.052±6.083
SPLIT-optimal	2.990±0.993	0.400±0.298	5.801±4.757	15.896±7.998
ReSPLIT	15.706±5.904	3.616±0.423	13.589±12.762	245.7±107.4

5.1 Experimental Setup

We use 12 publicly available classification datasets. Details are shown in Appendix B. All datasets use an 80/20 train–test split with stratification when feasible. For fairness mode, sensitive columns are removed before training. SPLIT-family models use binary features produced by the configured binarizer; the reported runs use GBDT stump threshold guessing with a maximum-threshold cap unless otherwise specified by the training script. Communities' target is binarized at the median.

We compare CART (Gini impurity, `max_depth=5`), SPLIT-greedy ($d_l = 2$, $d = 5$, $\lambda = 0.0001$, greedy Phase 2), SPLIT-optimal (same, optimal Phase 2), and ReSPLIT ($N = 30$ prefixes, $\varepsilon = 0.02$). SPLIT-greedy uses a top-50 feature cap in the evaluation script; SPLIT-optimal and ReSPLIT use all generated candidate features. The evaluation grid includes nofair, sample-calibrated fair

Table 3: Best average statistical-parity gap across sensitive attributes (mean±std over five random seeds; lower is better). "Base" denotes the uncalibrated prediction in fairness mode, "Sample" denotes label-aware sample-level post-processing, and "LPFR" denotes leaf-level Pareto fair recalibration. Parentheses report the model attaining the best average SP: S-g = SPLIT-greedy, S-o = SPLIT-optimal, ReS = ReSPLIT.

Dataset	Base (uncalibrated)	Sample (calibrated)	LPFR
Adult	0.1744±0.0116 (CART)	0.0026±0.0012 (CART)	0.1424±0.0169 (CART)
Bank	0.0281±0.0162 (ReS)	0.0024±0.0013 (S-g)	0.0072±0.0059 (S-g)
COMPAS	0.2248±0.0215 (ReS)	0.0013±0.0005 (S-g)	0.1554±0.0251 (CART)
German	0.0276±0.0285 (ReS)	0.0044±0.0021 (ReS)	0.0308±0.0310 (ReS)
Law School	0.0255±0.0062 (ReS)	0.0030±0.0005 (ReS)	0.0001±0.0003 (S-g)
ACSIncome	0.1671±0.0223 (S-g)	0.0017±0.0002 (S-o)	0.1181±0.0120 (CART)

mode, and LPFR mode. Metrics are test accuracy (%), training time (s), and SPD, DI, EOD as defined in Section 3.

5.2 Accuracy and Training Efficiency

Table 1 reports five-seed test accuracy. Accuracy is mixed across datasets: CART leads on Adult, Bank, German Credit, Spambase, Covertype, Thyroid, Communities, and ACSIncome; SPLIT-optimal leads on COMPAS; SPLIT-greedy leads on Law School; and ReSPLIT leads on HELOC and Heart. The largest positive margin for a SPLIT-family method appears on Heart, where ReSPLIT reaches $78.69 \pm 2.32\%$ versus CART's $76.39 \pm 2.49\%$. The largest negative margin appears on Covertype, where CART reaches $70.16 \pm 0.19\%$ and the SPLIT-family variants remain near 67%.

Table 2 reports mean training time. SPLIT-greedy is fastest on Adult, Bank, COMPAS, German Credit, HELOC, Spambase, Thyroid, and Communities, reflecting the effect of candidate-feature control. CART is fastest on Covertype, Heart, Law School, and ACSIncome. ReSPLIT is consistently slower because it trains many candidate trees; the gap is especially large on Adult, Covertype, and ACSIncome.

Figure 1 make the accuracy–time trade-off more explicit. SPLIT-greedy typically occupies the low-runtime region and is therefore the most attractive SPLIT-family option when candidate features are controlled. ReSPLIT occupies a different regime: it can recover accuracy on HELOC and Heart, but the cost is large on Adult, Covertype, and ACSIncome. The post-processing colors are often vertically close, indicating that fairness post-processing does not uniformly change accuracy; when accuracy moves, the direction depends on the dataset and post-processing method.

5.3 Fairness Without Calibration

Table 3 summarizes demographic-parity behavior by averaging SP over the sensitive attributes of each fairness dataset and reporting the best model for each post-processing regime. Without post-processing, sensitive-attribute removal alone is not enough to make all datasets fair: the best base SP is small on Bank, German Credit, and Law School, but remains high on Adult, COMPAS, and ACSIncome. This confirms that proxy features and dataset imbalance remain important after direct sensitive columns are removed.

5.4 Effect of Post-Training Calibration

Sample-level calibration nearly eliminates average SP on all six fairness datasets, with best average SP values between 0.0013 and 0.0044. This result should be interpreted as an optimistic diagnostic because the procedure uses labels to choose which predictions to flip. LPFR is more conservative but deployable: it is highly effective on Law School (best average SP 0.0001) and Bank (0.0072), improves COMPAS and ACSIncome relative to their uncalibrated base gaps, but does not close the gap on Adult. EO results remain mixed. For example, LPFR improves average EO on Bank, COMPAS, and Law School in the summary, whereas demographic-parity-oriented sample calibration can

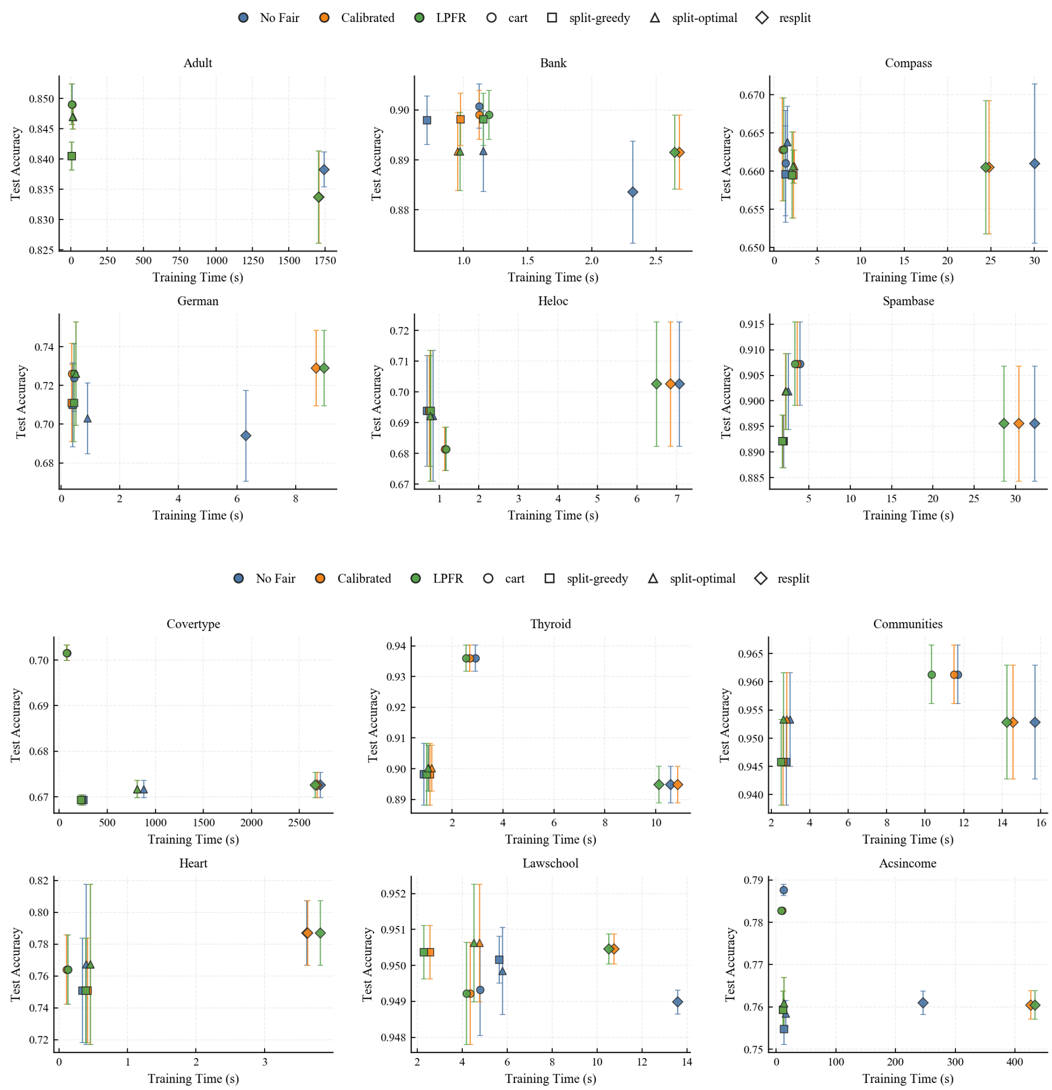

Figure 1: Test Accuracy v.s. Training Time on 12 benchmarks. Markers identify tree learners, colors identify post-processing modes, and error bars show 95% confidence intervals across seeds. The panel highlights that ReSPLIT can improve accuracy on selected small or medium datasets, but its computational cost grows substantially on larger datasets.

worsen EO on ACSIncome. In summary, SPLIT-family algorithms achieve accuracy competitive with CART on selected datasets but do not dominate uniformly; fairness behavior without post-processing is dataset-dependent; and the two calibration mechanisms expose a practical trade-off between optimistic gap removal and deployable tree-preserving recalibration.

Figure 2 illustrates why reporting multiple fairness metrics is necessary. On ACSIncome (RAC1P), the calibrated orange points nearly collapse SP to zero and push DI toward one, yet the EO panel remains far from zero. This is not a contradiction: demographic parity does not enforce equal opportunity. On Law School (race), by contrast, LPFR also moves SP and EO close to zero, showing that a deployable leaf-level adjustment can be effective when the learned tree structure already separates calibration-relevant regions.

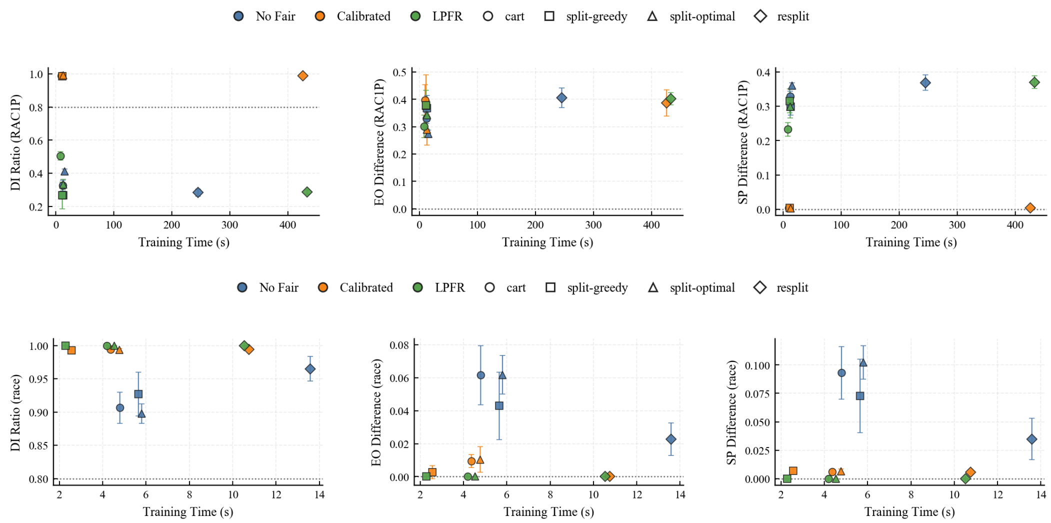

Figure 2: Fairness Metrics on ACSIncome (Race) and Law School (Race). ACSIncome illustrates the hard case: sample-level calibration drives SP and DI close to their demographic-parity targets, but EO remains high and LPFR is conservative. Law School illustrates the favorable case: both sample-level calibration and LPFR substantially reduce SP and EO while keeping DI near one.

ACKNOWLEDGEMENTS

We thank the authors of the SPLIT framework for releasing their open-source code, which served as the foundation for our fair-SPLIT implementation. We are also grateful to Xiangyu Zheng for her valuable guidance and insightful discussions for this project.

AUTHOR CONTRIBUTIONS

Ziyi Cheng conceived the project and formulated the core research idea. Cheng implemented the CART algorithm and drafted the initial manuscript of the final report. Kaidi Zha developed the SPLIT-series methods and designed the experimental evaluation. Both authors contributed to the fair-improved algorithm design, data analysis, and the writing and revision of the final report.

REFERENCES

Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. A reductions approach to fair classification. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2018.

Sina Aghaei, Mohammad Javad Azizi, and Phebe Vayanos. Learning optimal and fair decision trees for non-discriminative decision-making. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pp. 1418–1426, 2019. doi: 10.1609/aaai.v33i01.33011418.

Gaël Aglin, Siegfried Nijssen, and Pierre Schaus. Learning optimal decision trees using caching branch-and-bound search. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 3146–3153, 2020. doi: 10.1609/aaai.v34i04.5711.

Varun Babbar, Hayden McTavish, Cynthia Rudin, and Margo Seltzer. Near-optimal decision trees in a SPLIT second. In *Proceedings of the International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 2114–2175, 2025. doi: 10.48550/arXiv.2502.15988. URL `https://arxiv.org/abs/2502.15988`. ICML 2025 Spotlight.

Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. fairmlbook.org, 2017. https://fairmlbook.org.

L. Breiman, J. Friedman, R. Olshen, and C. Stone. *Classification and Regression Trees*. Wadsworth International Group, Belmont, CA, 1984.

Leo Breiman. Statistical modeling: The two cultures (with comments and a rejoinder by the author). *Statistical Science*, 16(3):199–231, 2001.

Ayman Chaouki, Jesse Read, and Albert Bifet. Branches: Efficiently seeking optimal sparse decision trees with AO*. In *Proceedings of the International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, 2025. URL https://arxiv.org/abs/2406.02175.

Somnath Basu Roy Chowdhury, Nicholas Monath, Ahmad Beirami, Rahul Kidambi, Avinava Dubey, Amr Ahmed, and Snigdha Chaturvedi. Enhancing group fairness in online settings using oblique decision forests. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. doi: 10.48550/arXiv.2310.11401. URL https://arxiv.org/abs/2310.11401. ICLR 2024 Spotlight.

André F. Cruz, Catarina Belém, Sérgio Jesus, João Bravo, Pedro Saleiro, and Pedro Bizarro. FairGBM: Gradient boosting with fairness constraints. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023. URL https://arxiv.org/abs/2209.07850.

Emir Demirović, Anna Lukina, Emmanuel Hebrard, Jeffrey Chan, James Bailey, Christopher Leckie, Kotagiri Ramamohanarao, and Peter J. Stuckey. MurTree: Optimal decision trees via dynamic programming and search. *Journal of Machine Learning Research*, 23(26):1–47, 2022. URL https://jmlr.org/papers/v23/20-520.html.

Houtao Deng, George Runger, and Eugene Tuv. Bias of importance measures for multi-valued attributes and solutions. In *International Conference on Artificial Neural Networks (ICANN)*, volume 6792 of *Lecture Notes in Computer Science*, pp. 293–300. Springer, 2011. doi: 10.1007/978-3-642-21738-8_38.

Jon Donnelly, Srikar Katta, Cynthia Rudin, and Edward Browne. The Rashomon importance distribution: Getting RID of unstable, single model-based variable importance. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Proceedings of the Innovations in Theoretical Computer Science Conference (ITCS)*, 2012.

Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian, Sonam Choudhary, Evan P. Hamilton, and Derek Roth. A comparative study of fairness-enhancing interventions in machine learning. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pp. 329–338, 2019. doi: 10.1145/3287560.3287589.

Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

Xiyang Hu, Cynthia Rudin, and Margo Seltzer. Optimal sparse decision trees. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 7265–7273, 2019.

Laurent Hyafil and Ronald L. Rivest. Constructing optimal binary decision trees is np-complete. *Information Processing Letters*, 5(1):15–17, 1976. ISSN 0020-0190. doi: https://doi.org/10.1016/0020-0190(76)90095-8.

Haewon Jeong, Hao Wang, and Flavio P. Calmon. Fairness without imputation: A decision tree approach for fair prediction with missing values. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 9558–9566, 2022. doi: 10.1609/aaai.v36i9.21189. URL https://arxiv.org/abs/2109.10431.

Nathanael Jo, Sina Aghaei, Jack Benson, Andrés Gómez, and Phebe Vayanos. Learning optimal fair classification trees: Trade-offs between interpretability, fairness, and accuracy. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2023. doi: 10.1145/3600211.3604664. URL `https://arxiv.org/abs/2201.09932`.

Faisal Kamiran and Toon Calders. Classifying without discriminating. In *Proceedings of the International Conference on Computer, Control and Communication (IC4)*, 2009.

Faisal Kamiran and Toon Calders. Data preprocessing techniques for classification without discrimination. *Knowledge and Information Systems*, 33(1):1–33, 2012.

Hector Kohler, Riad Akrour, and Philippe Preux. Breiman meets bellman: Non-greedy decision trees with MDPs. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025. doi: 10.1145/3711896.3736868. URL `https://arxiv.org/abs/2309.12701`.

Jimmy Lin, Chudi Zhong, Diane Hu, Cynthia Rudin, and Margo Seltzer. Generalized and scalable optimal sparse decision trees. In *Proceedings of the International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pp. 6150–6160, 2020.

Hayden McTavish, Chudi Zhong, Reto Achermann, Ilias Karimalis, Jacques Chen, Cynthia Rudin, and Margo Seltzer. Fast sparse decision tree optimization via reference ensembles. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 9604–9613, 2022. doi: 10.1609/aaai.v36i9.21194.

Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):115, 2021.

Aditya Krishna Menon and Robert C. Williamson. The cost of fairness in binary classification. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, volume 81 of *Proceedings of Machine Learning Research*, pp. 107–118, 2018.

Sreerama K. Murthy, Simon Kasif, and Steven Salzberg. A system for induction of oblique decision trees. *Journal of Artificial Intelligence Research*, 2(1):1–32, 1994. doi: 10.1613/jair.63.

Geoff Pleiss, Manish Raghavan, Felix Wu, Jon Kleinberg, and Kilian Q. Weinberger. On fairness and calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Edward Raff, Jared Sylvester, and Steven Mills. Fair forests: Regularized tree induction to minimize model bias. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 243–250. ACM, 2018. doi: 10.1145/3278721.3278742.

Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

Lesia Semenova, Cynthia Rudin, and Ronald Parr. On the existence of simpler machine learning models. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pp. 1827–1858, 2022. doi: 10.1145/3531146.3533232. arXiv preprint title: A Study of Rashomon Curves and Volumes: A New Perspective on Generalization and Model Simplicity in Machine Learning, arXiv:1908.01755.

Carolin Strobl, Anne-Laure Boulesteix, Achim Zeileis, and Torsten Hothorn. Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8:25, 2007.

Jacobus G. M. van der Linden, Mathijs M. de Weerdt, and Emir Demirović. Necessary and sufficient conditions for optimal decision trees using dynamic programming. *arXiv preprint arXiv:2305.19706*, 2023. URL `https://arxiv.org/abs/2305.19706`.

Sicco Verwer and Yingqian Zhang. Learning optimal classification trees using a binary linear program formulation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pp. 1625–1632, 2019. doi: 10.1609/aaai.v33i01.33011624.

Rui Xin, Chudi Zhong, Zhi Chen, Takuya Takagi, Margo Seltzer, and Cynthia Rudin. Exploring the whole Rashomon set of sparse decision trees. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Yongxin Yang, Irene Garcia Morillo, and Timothy M. Hospedales. Deep neural decision trees. *arXiv preprint arXiv:1806.06988*, 2018. URL `https://arxiv.org/abs/1806.06988`.

Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. Fairness constraints: A flexible approach for fair classification. *Journal of Machine Learning Research*, 20(75):1–42, 2019.

Richard Zemel, Yu Wu, Kevin Swersky, Toniann Pitassi, and Cynthia Dwork. Learning fair representations. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2013.

A METRIC DEFINITIONS

We provide formal definitions for all evaluation metrics referenced in the main text.

Gini Impurity (Breiman et al., 1984). For a dataset $\mathcal{D}$ with K classes, let C_i be the subset of $\mathcal{D}$ with label y_i. The Gini index is

$$\text{Gini}(\mathcal{D}) = 1 - \sum_{i=1}^{K} \left(\frac{|C_i|}{|\mathcal{D}|} \right)^2 . \tag{4}$$

At each internal node, CART selects the split minimizing the weighted child impurity:

$$s^* = \arg\min_{s \in \mathcal{S}} \ \frac{|\mathcal{D}_L|}{|\mathcal{D}|} \cdot \text{Gini}(\mathcal{D}_L) + \frac{|\mathcal{D}_R|}{|\mathcal{D}|} \cdot \text{Gini}(\mathcal{D}_R). \tag{5}$$

Statistical Parity Difference (SPD) (Dwork et al., 2012; Barocas et al., 2017). Given protected attribute A partitioning the data into groups G and hard predictions $\hat{Y} \in \{0, 1\}$,

$$\text{SPD} = \max_{g \in G} P(\hat{Y} = 1 \mid A = g) \ - \ \min_{g \in G} P(\hat{Y} = 1 \mid A = g). \tag{6}$$

SPD = 0 indicates perfect demographic parity; values above 0.1 are conventionally considered unfair.

Disparate Impact (DI) (Barocas et al., 2017).

$$\text{DI} = \frac{\min_{g \in G} \ P(\hat{Y} = 1 \mid A = g)}{\max_{g \in G} \ P(\hat{Y} = 1 \mid A = g)}. \tag{7}$$

DI = 1 is ideal; DI < 0.8 conventionally indicates adverse impact per the four-fifths rule.

Equal Opportunity Difference (EOD) (Hardt et al., 2016).

$$\text{EOD} = \max_{g \in G} P(\hat{Y} = 1 \mid Y = 1, A = g) \ - \ \min_{g \in G} P(\hat{Y} = 1 \mid Y = 1, A = g). \tag{8}$$

EOD = 0 indicates equal true positive rates across all groups.

All metrics are estimated empirically on the test set. Groups with fewer than 50 test samples are excluded.

B DETAILS OF DATASETS

We evaluate on 12 benchmark datasets (Table 4). Six include sensitive attributes for fairness evaluation: Adult (sex, race), Bank (marital status, age), COMPAS (sex, race), German Credit (personal status, age), Law School (race, gender), and ACSIncome (sex, race). The remaining—HELOC, Spambase, Covertype, Thyroid, Communities, and Heart—are used for accuracy and efficiency comparisons; Covertype is multiclass, while Communities is binarized at the median target value.

Table 4: Dataset summary. N: total samples; d: number of features. Sensitive attributes are listed where applicable.

Dataset	N	d	Task	Sensitive Attributes
Adult	48,842	14	binary	Sex, Race
Bank	45,211	16	binary	Marital, Age
COMPAS	18,834	13	binary	Sex, Race
German	1,000	20	binary	Personal Status, Age
HELOC	10,000	23	binary	—
Spambase	4,601	57	binary	—
Covertype	581,012	54	7-class	—
Thyroid	7,200	21	binary	—
Communities	1,994	127	binary	—
Heart	303	13	binary	—
Law School	20,580	11	binary	Race, Gender
ACSIncome	195,666	10	binary	Sex, Race

C Experiments Visualizations

This appendix includes the remaining fairness panels generated from the same five-seed summary used in the main text. Each panel reports DI, EO, and SP against training time for one dataset–attribute pair.

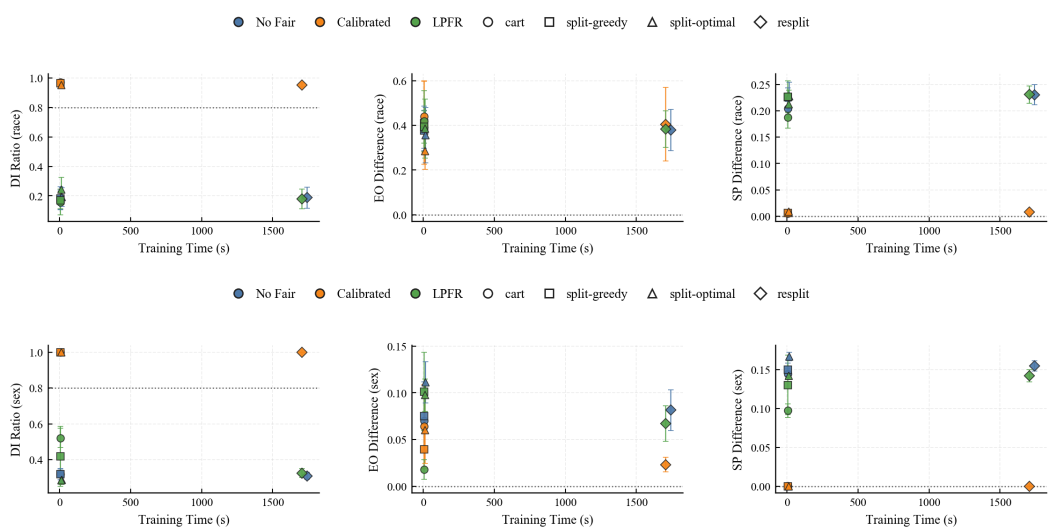

Figure 3: Fairness Metrics on Adult (Race) and Adult (Sex).

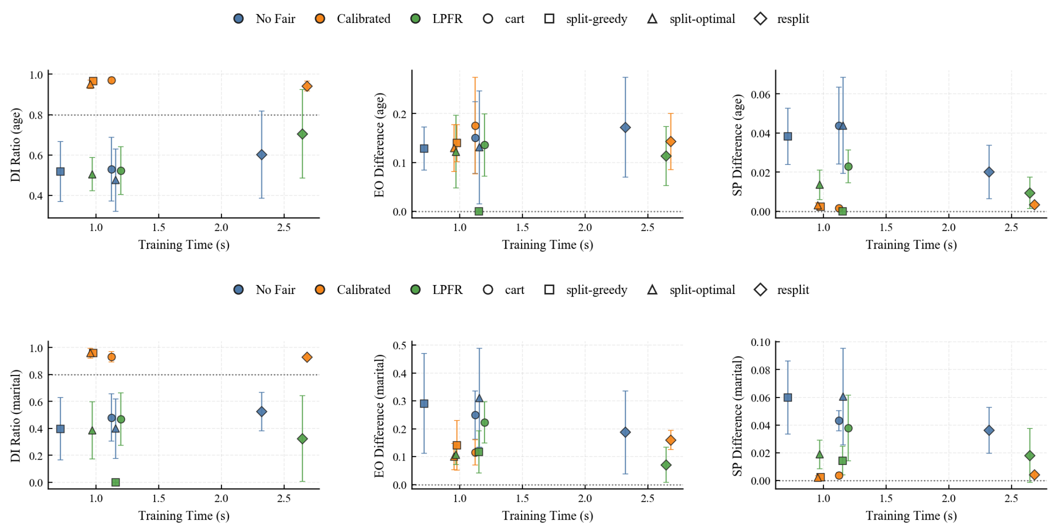

Figure 4: Fairness Metrics on Bank (Age) and Bank (Marital Status).

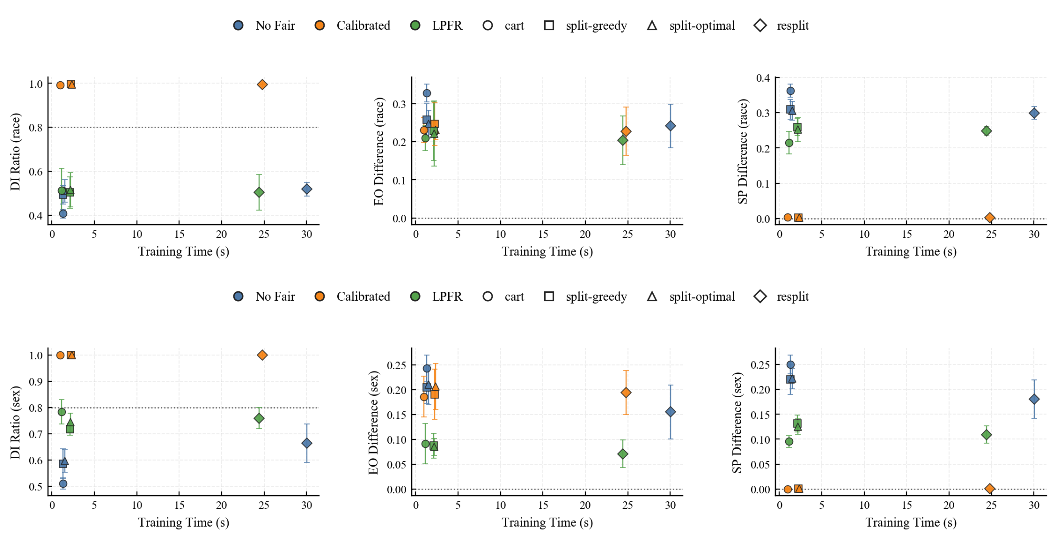

Figure 5: Fairness Metrics on COMPAS (Race) and COMPAS (Sex).

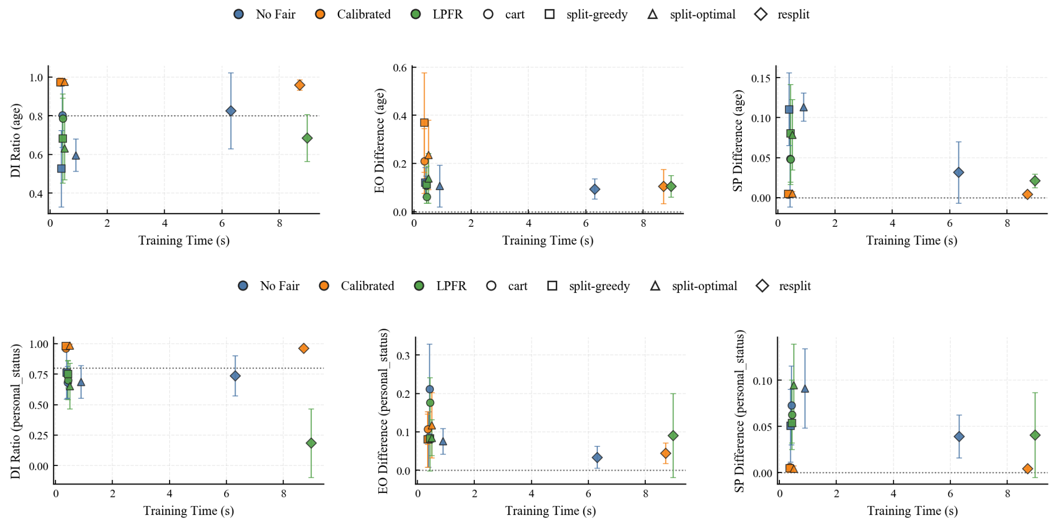

Figure 6: Fairness Metrics on German Credit (Age) and German Credit (Personal Status).

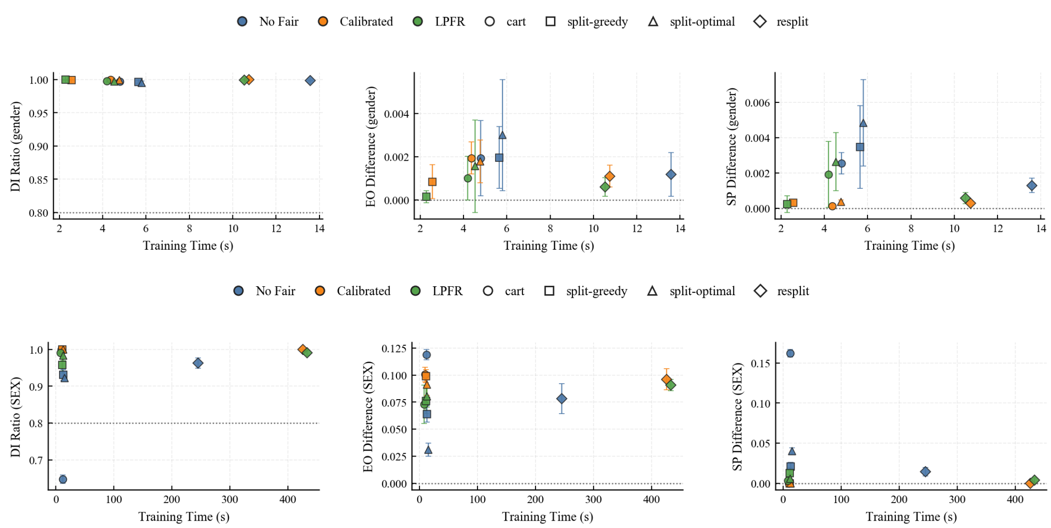

Figure 7: Fairness Metrics on Law School (Gender) and ACSIncome (Sex).